

THE IMBEWU INVESTIGATION

Week 1 — Discovery & Hypothesis Building

1. Stakeholder Brief

Sizwe Khumalo — Head of Sales — Monday 09:14

"i've been looking at the latest revenue report and something's off in Western Cape.

Numbers are down compared to last year, but our store visit reports say foot traffic is roughly flat. That doesn't add up.

I have an exec readout in three weeks and I need to walk in there knowing what's going on.

I want to know what's happening, why, and what we should do about it."

Key phrase: "foot traffic is roughly flat." This tells us customers are still walking through the door — but spending less per visit, buying fewer items, or shifting to cheaper products. The investigation must explain the revenue-per-visit gap, not overall footfall.

2. Data Overview

Six tables covering January 2024 – June 2025 across 45 stores nationwide.

Table	Rows	Key Fields
stores	45	store_id, format, province, suburb
products	48	product_id, category, sub_category, unit_price
customers	3,000	customer_id, loyalty_tier (Bronze / Silver / Gold)
transactions	9,164	transaction_id, store_id, customer_id (nullable), transaction_date
transaction_items	48,641	item_id, transaction_id, product_id, quantity, unit_price_at_sale, discount_applied
promotions	4	promo_id, promo_name, category_targeted, start_date, end_date, discount_pct

2.1 Data Quality Issues Found

- stores.province — inconsistent casing ("Western Cape" and "western cape", "Gauteng" and "gauteng"). Fix: normalise to Title Case before all grouping queries.
- stores.store_manager — 1 NULL. Not material for revenue analysis.
- transactions.customer_id — 3,911 NULLs (42.7%). Expected: these are anonymous walk-in customers not enrolled in the loyalty programme. Include in revenue totals; exclude from loyalty-tier analysis.
- customers.gender — 139 NULLs (4.6%). Not used in primary analysis.
- customers.birth_year — 161 NULLs (5.4%). Not used in primary analysis.

2.2 Year-Over-Year Comparison Note

Important — Like-for-Like Period

The dataset contains full-year 2024 but only H1 2025 (January–June).
All YoY comparisons are restricted to H1 (Jan–Jun) in both years to ensure a fair comparison.
Full-year 2024 data is used only for trend and context queries.

3. Investigation Hypotheses

Based on profiling the data and re-reading the stakeholder brief, four testable hypotheses have been identified in order of suspected impact:

#	Hypothesis	What we'd expect to see	Query approach
A	Mega-format stores in WC are underperforming, pulling the provincial average down	Revenue down in Mega format in WC; Express/Market flat or up. Nationally Mega stores also struggling.	Revenue by store format, WC vs national; store-level breakdown
B	Product mix shift — customers buying lower-value categories, reducing average basket	Groceries and Household (high volume, lower margin) shrinking in WC; higher-value categories flat.	Category revenue by province YoY; WC share of category vs national
C	Silver-tier loyalty customers in WC are spending less or visiting less often	Silver revenue and transaction count down in WC; Gold and Bronze flat. Basket size decline concentrated in Silver.	Revenue and visit count by loyalty tier, WC vs national
D	No active promotions in WC during the decline period, while other provinces benefitted	Promotion activity concentrated outside WC in H1 2025; WC receives fewer promo discounts.	Promotion coverage by province and date; discount volume by province

4. Week 1 Profiling Findings

Initial profiling queries confirm the problem is real and WC-specific:

Province	H1 2024 Revenue	H1 2025 Revenue	Change
Western Cape	R388,788	R365,245	-6.0%
Gauteng	R251,827	R274,484	+9.0%
KwaZulu-Natal	R218,302	R249,409	+14.3%
Eastern Cape	R88,529	R97,117	+9.7%

- WC is the ONLY province with declining H1 revenue. Every other province grew 9–14%. This is a WC-specific issue, not a national trend.
- WC transaction count (foot traffic proxy) dropped only 2.3% — confirming Sizwe's observation that visits are roughly stable.
- WC average basket fell from R387.62 to R372.70 (-3.9%). Nationally, every province's average basket grew. The problem is spend-per-visit, not customer count.

5. Week 2 Investigation Plan

The following lines of inquiry will be pursued in Week 2 in priority order:

Pri	Investigation	Key SQL
1	Confirm Mega-store drag (H-A). Break WC revenue by format and then by individual store. Compare against national Mega performance.	Revenue grouped by format + province; store-level YoY; Mega category breakdown
2	Test product mix shift (H-B). Compare WC category revenue YoY vs national. Identify which categories are losing share.	Category revenue WC vs national, H1 2024 vs H1 2025
3	Test Silver-tier erosion (H-C). Revenue, visit count, and basket size by loyalty tier. Monthly trend to identify when decline started.	Tier revenue + visit count YoY; monthly Silver trend; basket by tier
+	Pap Power Promo — assess volume uplift vs revenue impact for Naledi (Marketing)	Maize Meal units + revenue April 2024 vs 2025; discount cost vs revenue gain; basket halo effect

Hypotheses D (promo coverage gap) will be held in reserve and tested only if A, B, and C do not fully explain the revenue decline.